



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 01

Team No.:

Project Title: CPU SCHEDULING SIMULATOR

Team Members

Roll Number	Student Name	Signature
2520030017	G.VARSHINI	Varshini
2520030430	K.PRAGNYA	Pragnya
2520030482	M.AVISHYA	Avishya

1. Abstract

The **CPU Scheduling Simulator** is a software application designed to demonstrate and compare different CPU scheduling algorithms used by operating systems. The system takes process details such as Process ID, Arrival Time, Burst Time, and Priority as input and simulates algorithms such as FCFS, SJF, Priority Scheduling, and Round Robin. It calculates important performance measures including Waiting Time, Turnaround Time, and Average Waiting Time. A Gantt Chart is generated to visually represent the order and duration of process execution. The simulator helps students understand how the CPU selects and manages processes and provides an easy way to compare the performance of different scheduling algorithms. The project can be implemented using C/C++ on Linux with POSIX concepts and system calls.

2. Problem Statement

CPU scheduling is an important function of an operating system. It decides which process should get the CPU and for how long. Different scheduling algorithms give different results in terms of waiting time, turnaround time, and response time.

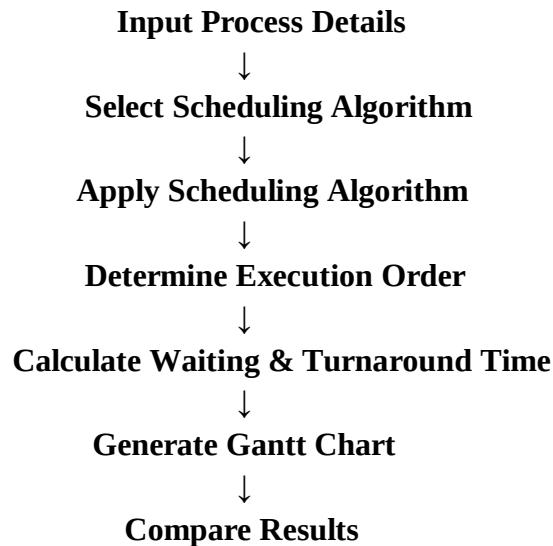
The CPU Scheduling Simulator will simulate different CPU scheduling algorithms and show how processes are executed. It will help users understand and compare scheduling techniques without needing to run actual operating-system processes.

3. Objectives

1. To understand the concept of CPU scheduling.

2. To simulate different scheduling algorithms.
3. To calculate waiting time and turnaround time.
4. To compare the performance of different algorithms.
5. To display the execution order of processes.
6. To generate a Gantt chart for visualization.
7. To understand how the operating system manages CPU resources.

4. Proposed Methodology



Steps:

1. Take process information such as **Process ID, Arrival Time, and Burst Time**.
2. Allow the user to select a scheduling algorithm.
3. Apply the selected algorithm to determine which process runs first.
4. Simulate CPU execution.
5. Calculate waiting time and turnaround time for each process.
6. Display the process execution sequence using a Gantt chart.
7. Calculate average waiting time and average turnaround time.
8. Compare the results of different scheduling algorithms.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
C / C++	Implement the simulator
Linux OS	Development and execution environment
fork()	Create child processes if process creation is demonstrated
wait() / waitpid()	Wait for process completion
getpid()	Get process ID
sleep() / usleep()	Simulate process execution time

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030017	G.VARSHINI	Scheduling algorithms & calculations
2520030430	K.PRAGNYA	Linux/POSIX & process simulation
2520030482	M.AVISHYA	Gantt chart, output & documentation

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system for development and execution
C / C++	Main programming language
GCC Compiler	Compiling and running the program
VS Code	Writing and editing source code
Git & GitHub	Storing and managing project source code

8. Expected Outcome

- Successfully simulate different CPU scheduling algorithms.
- Display the order in which processes are executed.
- Generate a clear Gantt chart.
- Calculate waiting time and turnaround time.
- Calculate average scheduling performance.
- Allow users to compare different algorithms.
- Provide a simple way to understand how CPU scheduling works in an operating system.

9. GitHub Repository

GitHub Repository Link: https://github.com/pragnya2202/CPU_Scheduling_OSSP

https://github.com/Avishya2008/CPU_Scheduling_OSSP

https://github.com/varshinigeriganti24/CPU_Scheduling_OSSP

Repository Created: Yes

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: Dr. D.Bheekya

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____